Improving Reliability of Multi-Cell MIMO Mission Critical Communication Systems using LDPC in 5G

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Multi-Cell MIMO System Model

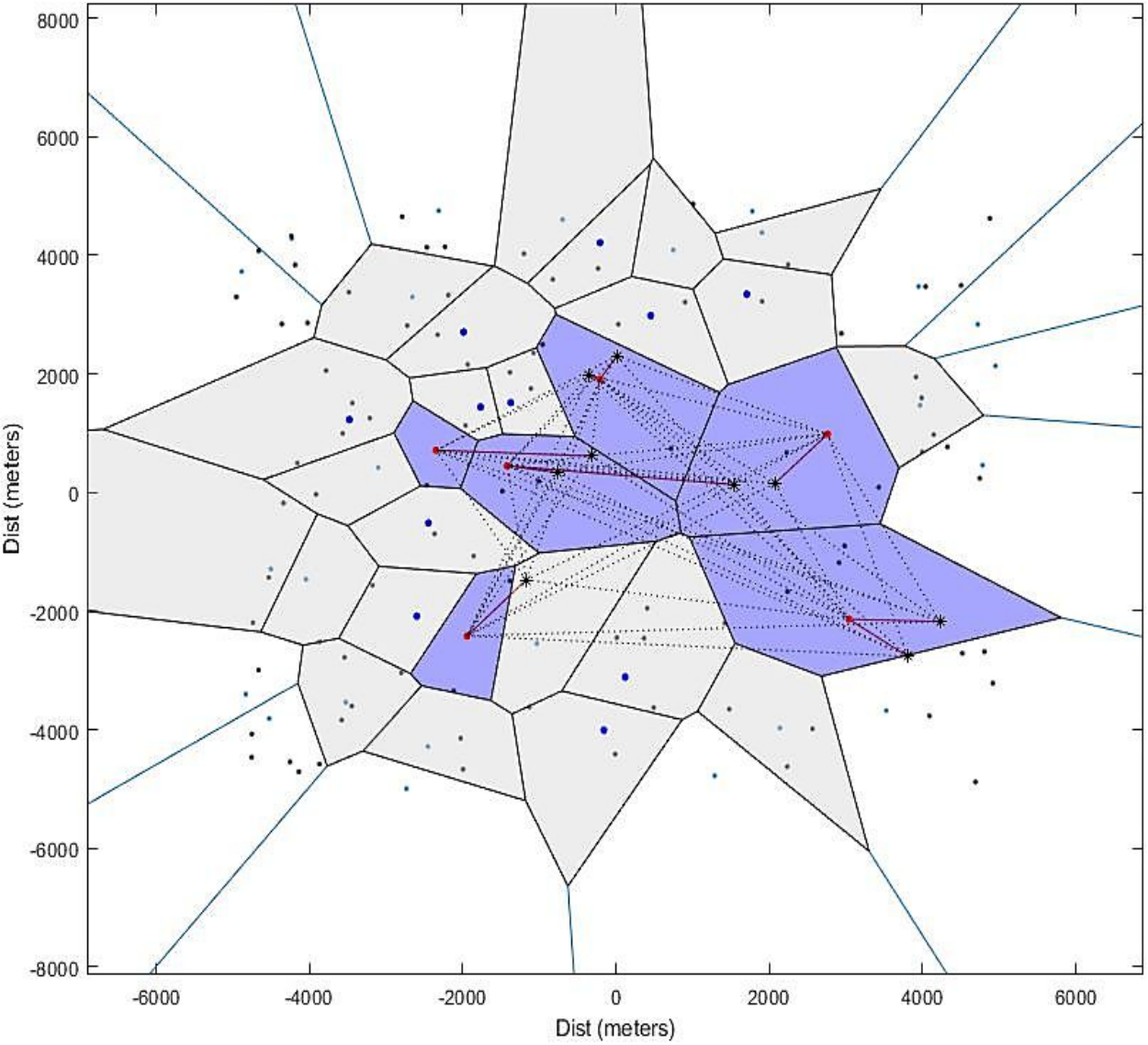


Figure 1: A Multi-Cell MIMO system. Red dots (●) are the eNBs in MC configuration and their cell area is shaded in blue, MCC UEs are represented by an asterisk (*), while regular UEs are represented by simple black dots (●) and all other eNBs are represented by Blue dots (●)

Multi-Cell MIMO System Model

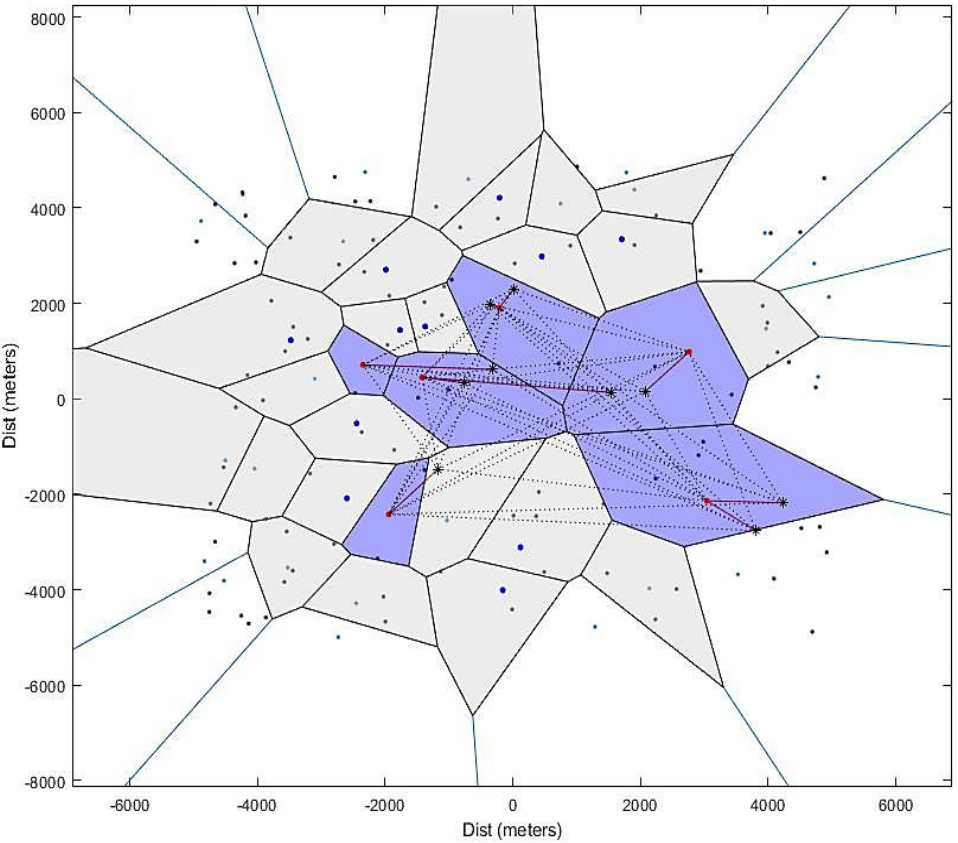


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- Signals from the multiple eNBs are coherently combined => Higher SNR => Lower BER.
- MC-MIMO systems display tremendous potential for group communication and MCC applications.
- How are eNBs chosen ?
- Transceiver at UEs minimise Group MMSE [1].

[1] . Jagyasi, A. Daher, and M. Coupechoux, "Multi-cell mimo transceiver design for mission-critical communication," 2019.

Mission Critical Systems

- Mission Critical systems impose stringent on Key Performance Indicators (KPIs) like [2]:

1. Ultra-Reliability: BER $< 10^{-10}$.
2. Ultra-Low Latency: Round Trip Time (RTT) $< 1ms$.
3. Availability: Availability refers to the system's endurance against possible outage scenarios.
   $P_{out} < 10^{-9}$.

[2] N. A. Mohammed, A. M. Mansoor, and R. B. Ahmad, "Mission-critical machine-type communications: An overview and perspectives towards 5g," IEEE Access, 2019.

Mission Critical Systems

- Role of Reliability in Mission Critical Communication Systems is two fold:
  1. To minimise Errors.
  2. To limit or completely eliminate retransmission.

- Latency Target: $< 1ms$ Round Trip Latency.

Understanding LDPC Codes

- Low-Density Parity Check (LDPC) codes are a class of **linear block codes** and they are reminiscent of a sparse Parity Check Matrix (PCM) $\boldsymbol{H}$. LDPC codes are of two types [3]:

  1. Regular codes – are codes with constant ω_c and ω_r.
  2. Irregular codes – have no such constraint on ω_c and ω_r.

  where ω_c - column hamming weight, ω_r - row hamming weight respectively.

- Irregular codes allow for better performance, due to their relaxed constraints.

[3] Tom Richardson and Ruediger Urbanke. Modern coding theory. Cambridge university press, 2008.

Regular LDPC code

$$\boldsymbol{H} = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

Irregular LDPC code

$$\boldsymbol{H} = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

- An LDPC code is fully specified by its PCM .

LDPC in 5G

- The 5G standard has adopted LDPC codes as the channel coding scheme of choice, for downlink and uplink data channels [4] (as of 3GPP Release 15) due to their formidable performance and flexibility.

- The 5G standard has 2 sets of 51 PCMs each defined, aka, Base Graph 1 and Base Graph 2 with code rates $\left(\frac{1}{3}\right)$ and $\left(\frac{1}{5}\right)$ respectively. That is a total of 102 PCMs.

- Packet sizes supported by Base Graph 1: [44 – 8448] bits,
  Base Graph 2: [20 – 3840] bits,

- Note: LDPC codes are **Linear Block Codes**. 102 PCMs => 102 discrete packet sizes in between the ranges mentioned above.

[4] Stefan Parkvall, Erik Dahlman, Anders Furuskar, and Mattias Frenne. Nr: The new 5g radio access technology. IEEE Communications Standards Magazine, 1(4):24–30, 2017.

- 3GPP Technical Documentation.

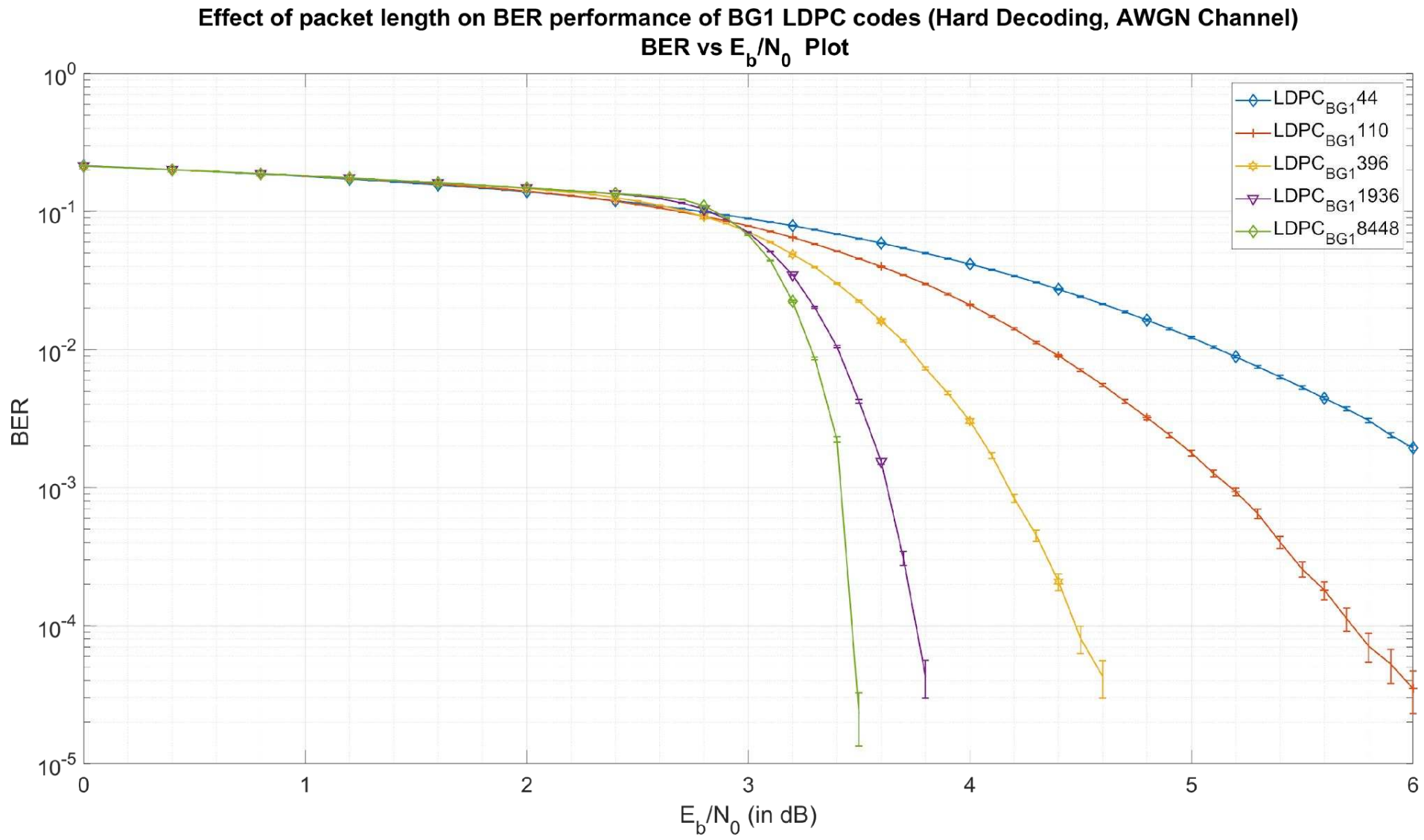
ETSI TS 138 212 V15.2.0 (2018-07)

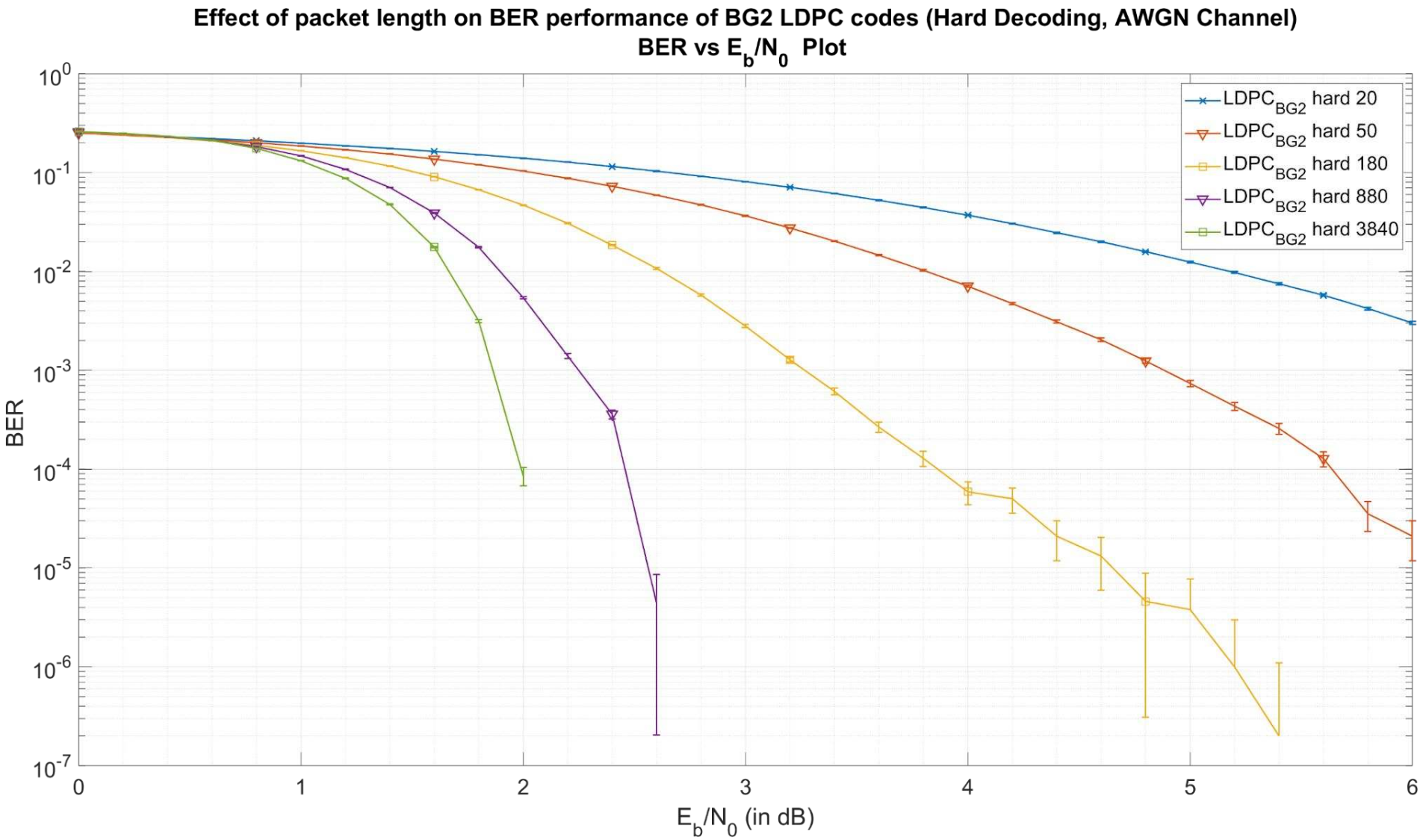


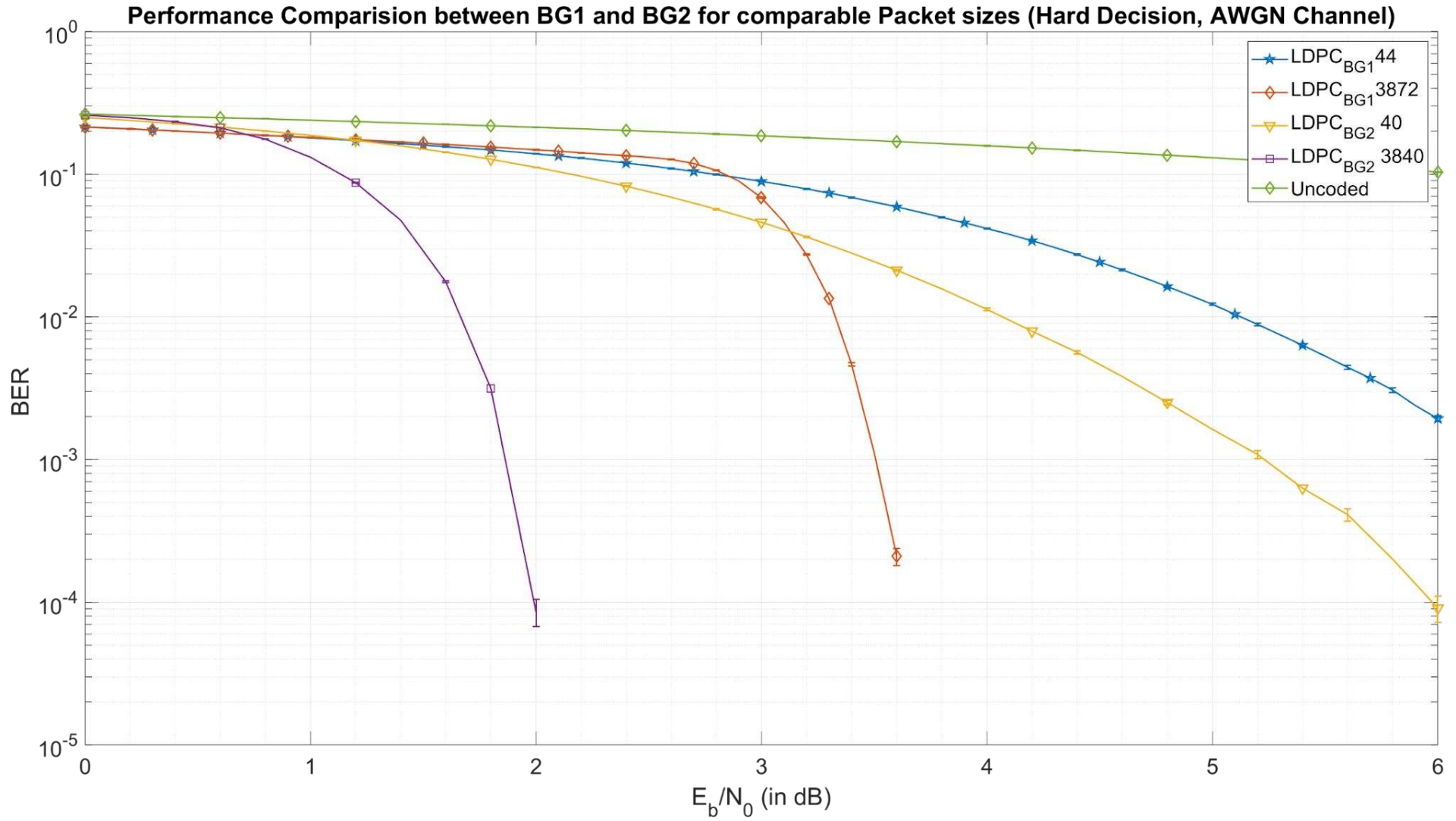
5G;
NR;
Multiplexing and channel coding
(3GPP TS 38.212 version 15.2.0 Release 15)

Performance of LDPC in 5G

- All simulations are performed in an AWGN channel for Hard decision decoding and Soft decision decoding, using MMSE receive filter (optimal).
- The error bars indicate the 95% confidence interval (the uncertainty arises due to the finite number of samples).


Effect of packet length on BER performance of BG1 LDPC codes (Hard Decoding, AWGN Channel)
BER vs Eb/N0 Plot
BER
Eb/N0 (in dB)
LDPCBG1 44
LDPCBG1 110
LDPCBG1 396
LDPCBG1 1936
LDPCBG1 8448


Effect of packet length on BER performance of BG2 LDPC codes (Hard Decoding, AWGN Channel)
BER vs Eb/N0 Plot
LDPCBG2 hard 20
LDPCBG2 hard 50
LDPCBG2 hard 180
LDPCBG2 hard 880
LDPCBG2 hard 3840
BER
Eb/N0 (in dB)
10^0
10^-1
10^-2
10^-3
10^-4
10^-5
10^-6
10^-7
0
1
2
3
4
5
6


Performance Comparision between BG1 and BG2 for comparable Packet sizes (Hard Decision, AWGN Channel)
BER
E_b/N_0 (in dB)
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
1
2
3
4
5
6
$LDPC_{BG1}$ 44
$LDPC_{BG1}$ 3872
$LDPC_{BG2}$ 40
$LDPC_{BG2}$ 3840
Uncoded

Performance of LDPC in 5G

- Observations:
  - Base Graph 2 > Base Graph 1, and
  - Larger Pkt Size > Smaller Pkt Size.

- LDPC packet sizes play a pivotal role in determining performance of the code.

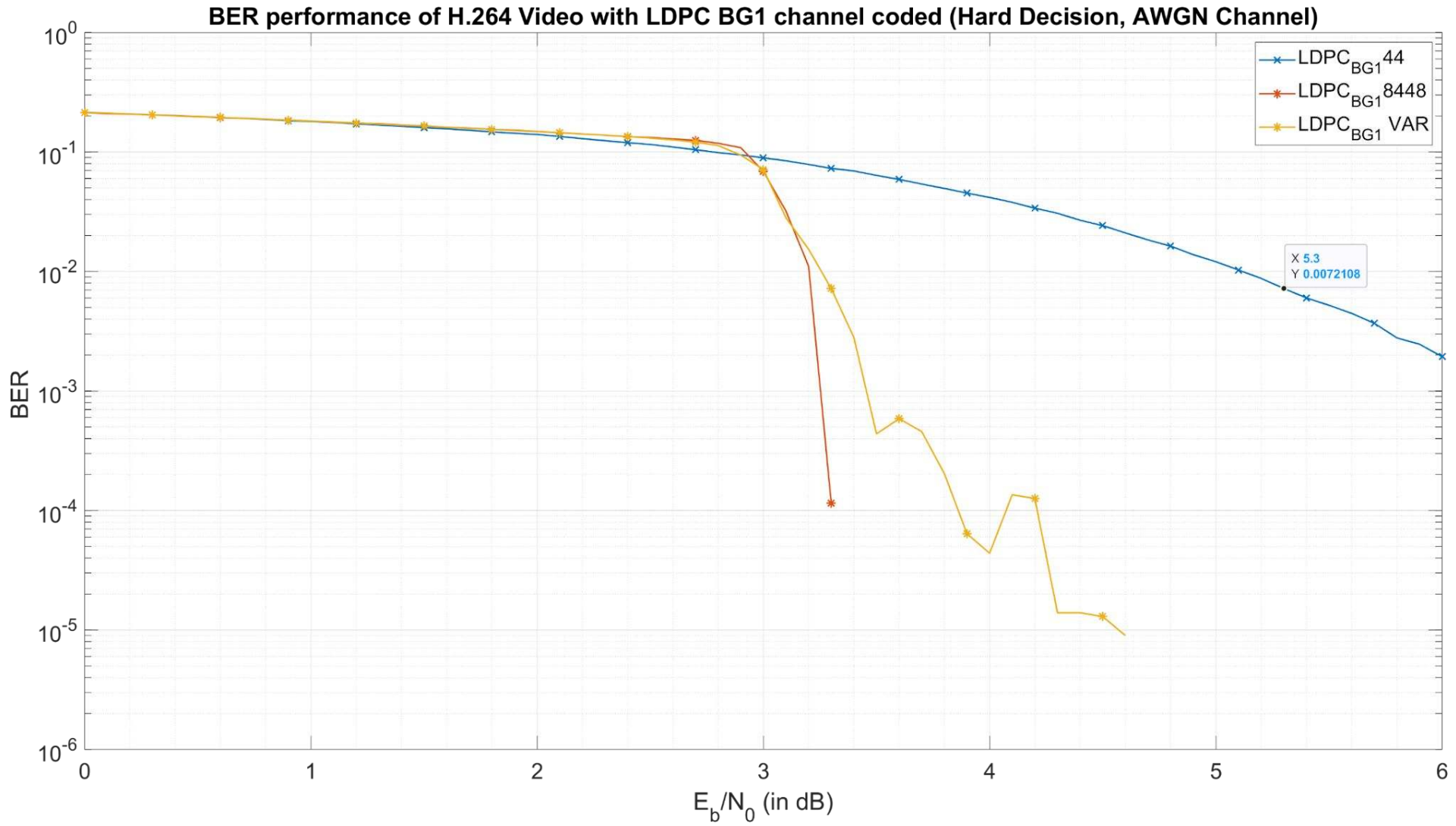
- Therefore from above, segmenting the input data into PktS of 8448 for BG1 and 3840 for BG2, low BER can be achieved.

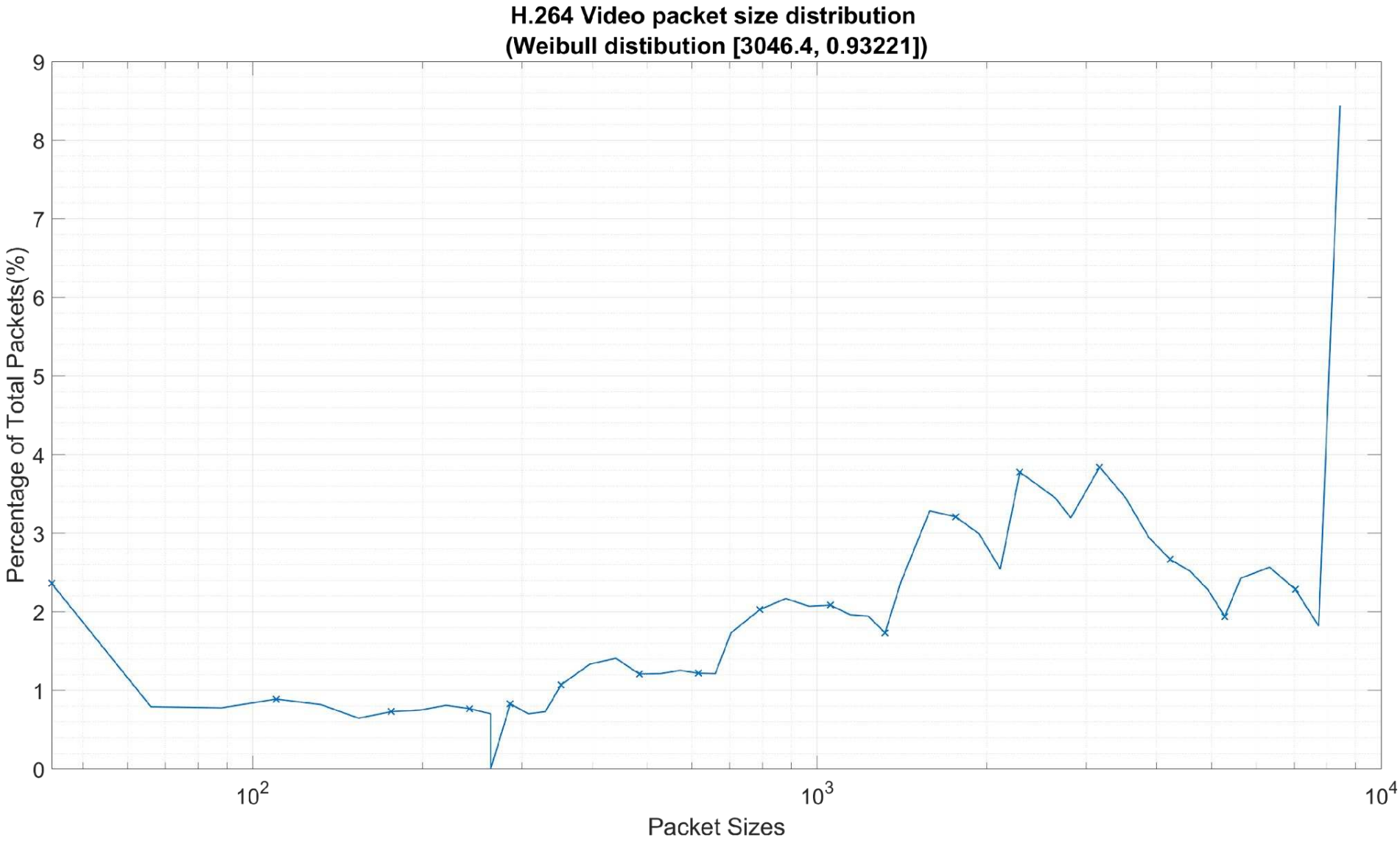
- However, the above policy is only applicable to non real-time systems as the data needs to be available apriori. For real-time systems, especially, MCC systems, the data is generated on the fly in real-time.

Performance of video using LDPC

- The distribution of packet sizes of a H.264 encoded video is found to be best explained by a Weibull distribution [5], for video lectures and video conferencing applications.

[5] Khaled Salah, Fahd Al-Haidari, MH Omar, and A Chaudhry. Statistical analysis of h. 264 video frame size distribution. IET communications, 5(14):1978–1986, 2011.


BER performance of H.264 Video with LDPC BG1 channel coded (Hard Decision, AWGN Channel)
LDPC BG1 44
LDPC BG1 8448
LDPC BG1 VAR
X 5.3
Y 0.0072108
BER
Eb/N0 (in dB)
10^0
10^-1
10^-2
10^-3
10^-4
10^-5
10^-6
0
1
2
3
4
5
6

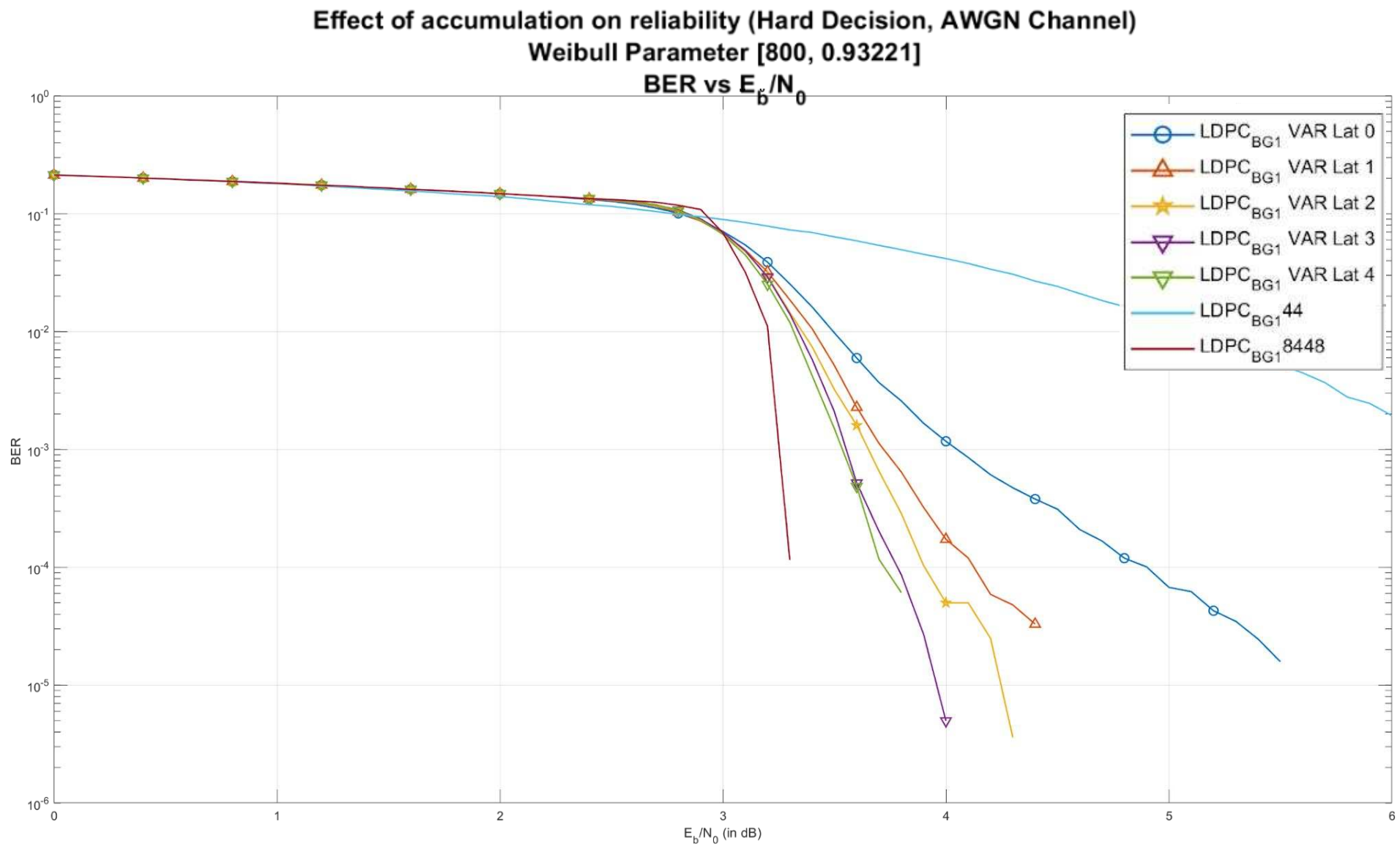

H.264 Video packet size distribution
(Weibull distibution [3046.4, 0.93221])
Percentage of Total Packets(%)
0
1
2
3
4
5
6
7
8
9
10^2
10^3
10^4
Packet Sizes

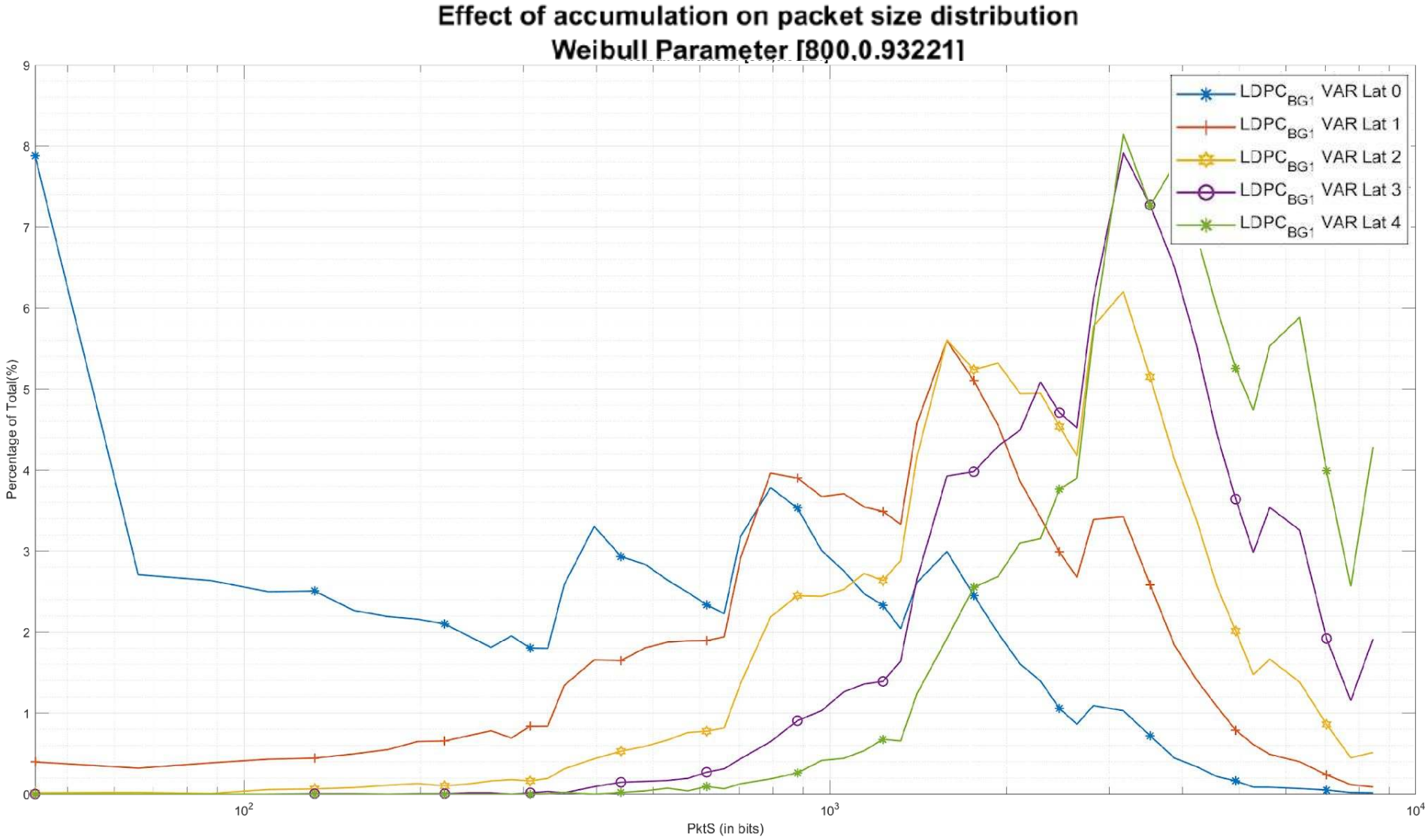
Performance of video using LDPC

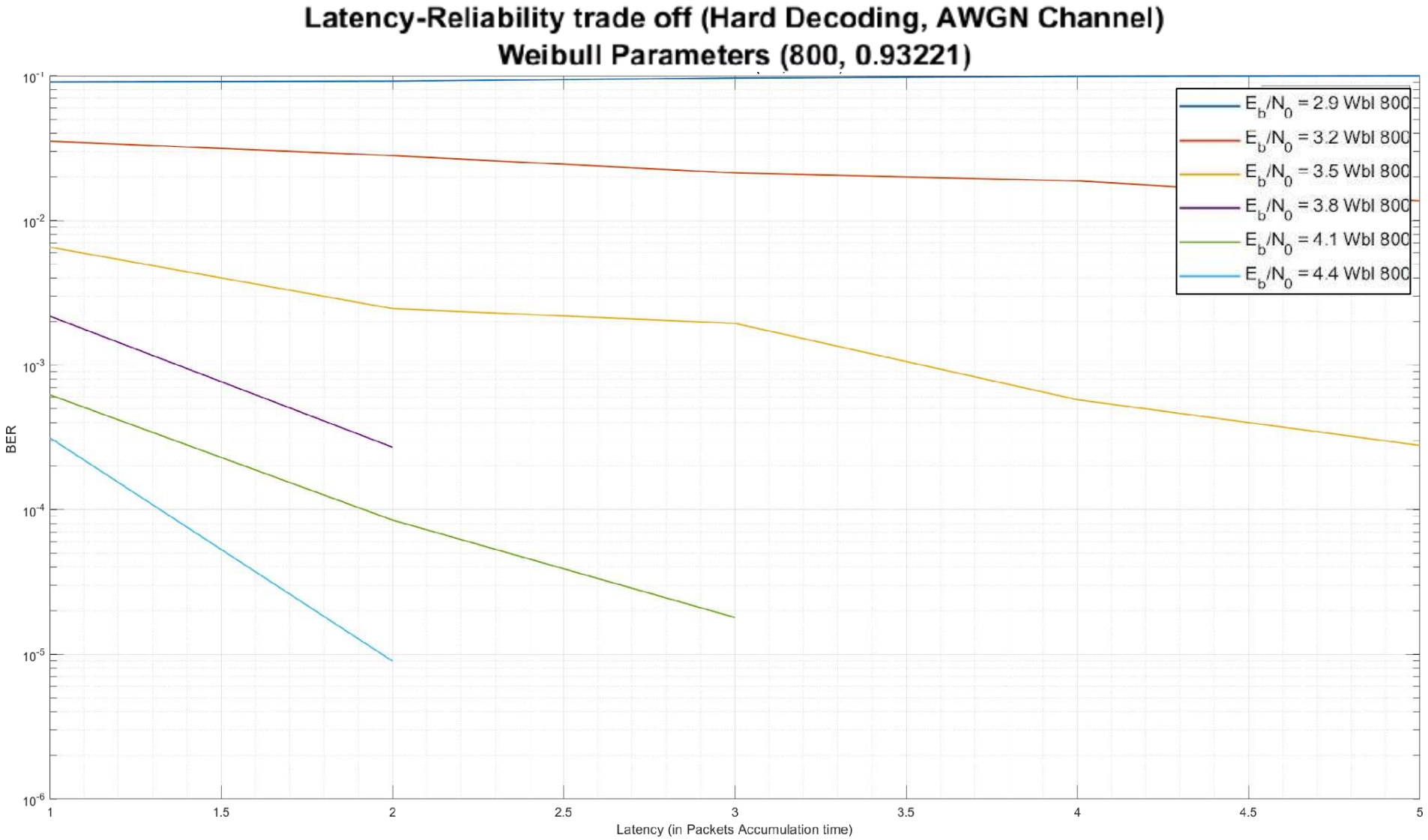
- Why H.264 video ?
- H.264 video compression standard is chosen for analysis as it is modern and the de-facto standard for most practical applications.

Latency-Reliability Trade-off

- To combat the degradation of real-time system we propose accumulation (i.e. buffering) of the input packets.


Effect of accumulation on reliability (Hard Decision, AWGN Channel)
Weibull Parameter [800, 0.93221]
BER vs Eb/N0
BER
Eb/N0 (in dB)
LDPCBG1 VAR Lat 0
LDPCBG1 VAR Lat 1
LDPCBG1 VAR Lat 2
LDPCBG1 VAR Lat 3
LDPCBG1 VAR Lat 4
LDPCBG1 44
LDPCBG1 8448


Effect of accumulation on packet size distribution
Weibull Parameter [800,0.93221]
LDPC BG1 VAR Lat 0
LDPC BG1 VAR Lat 1
LDPC BG1 VAR Lat 2
LDPC BG1 VAR Lat 3
LDPC BG1 VAR Lat 4
Percentage of Total(%)
PktS (in bits)
0
1
2
3
4
5
6
7
8
9
10^2
10^3
10^4


Latency-Reliability trade off (Hard Decoding, AWGN Channel)
Weibull Parameters (800, 0.93221)
BER
Latency (in Packets Accumulation time)
Eb/N0 = 2.9 Wbl 800
Eb/N0 = 3.2 Wbl 800
Eb/N0 = 3.5 Wbl 800
Eb/N0 = 3.8 Wbl 800
Eb/N0 = 4.1 Wbl 800
Eb/N0 = 4.4 Wbl 800

Latency-Reliability Trade-off

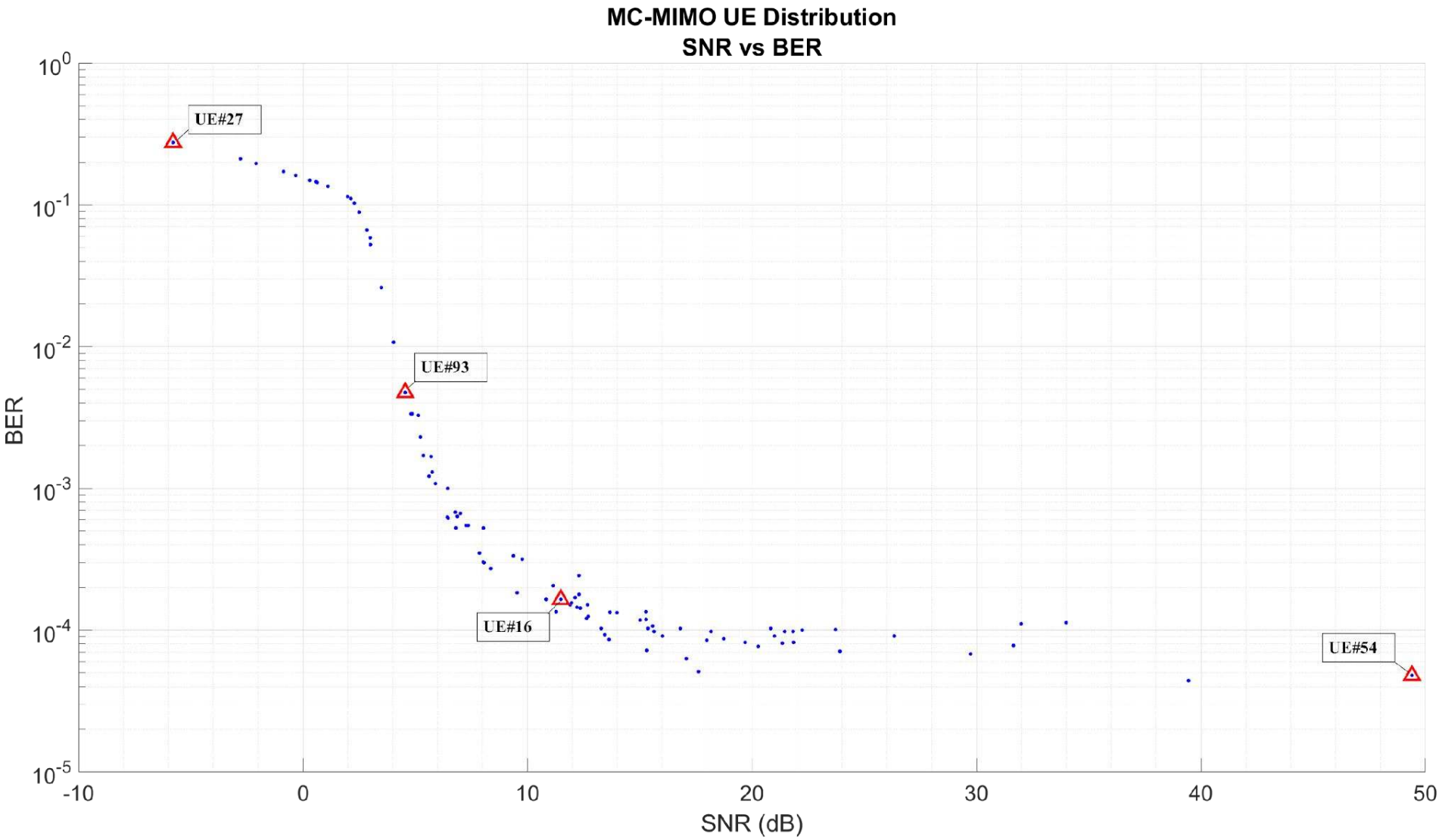
- Accumulation introduces a delay.

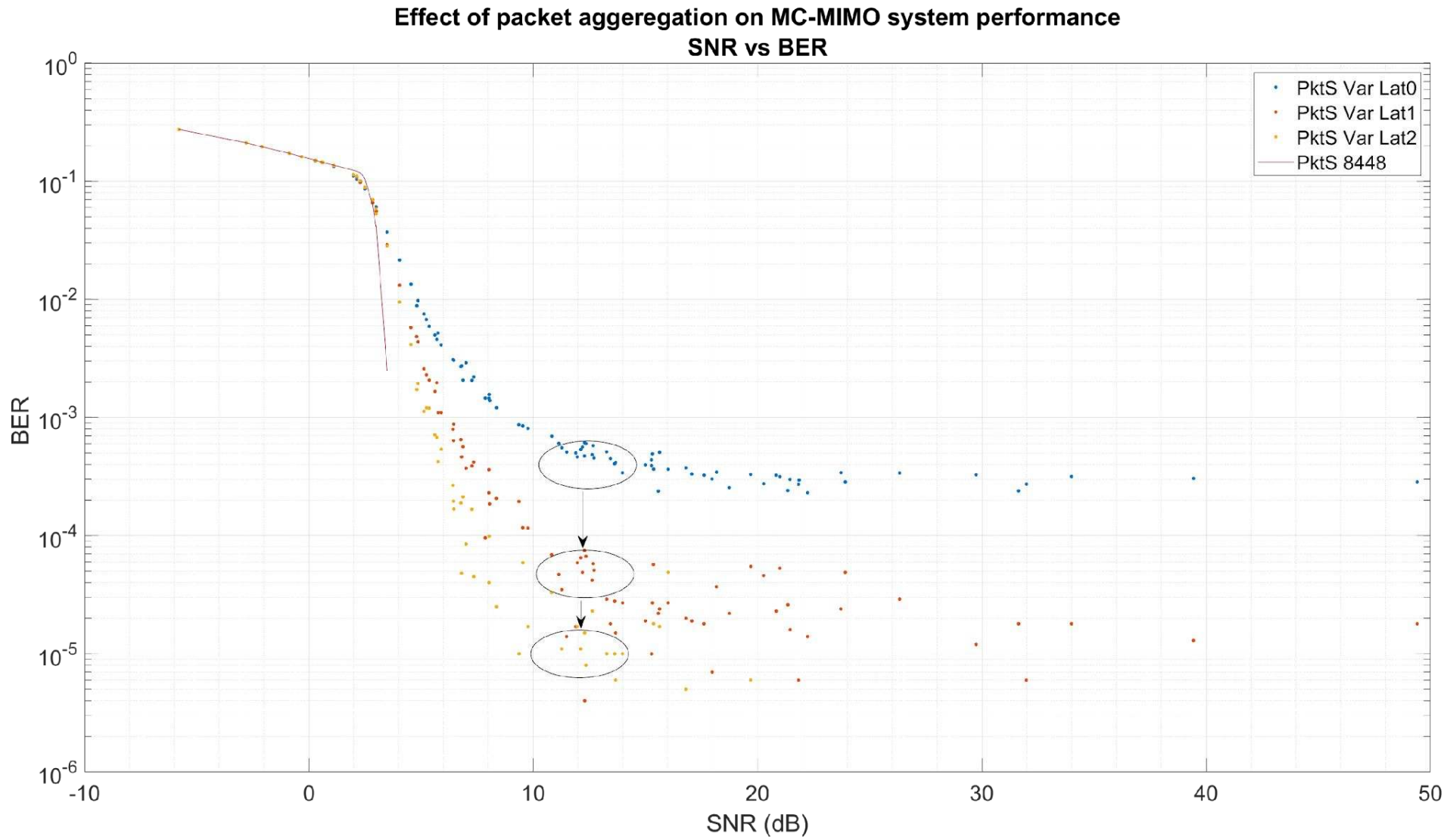
- Observation:
  - Higher Accumulation => Performance closer to Non-Realtime System.
  - The Latency-Reliability trade-off is steeper at higher $\frac{E_b}{N_0}$.

Performance of Video over a MC-MIMO System

Video Stream as received by UEs




MC-MIMO UE Distribution
SNR vs BER
BER
SNR (dB)
UE#27
UE#93
UE#16
UE#54
10^0
10^-1
10^-2
10^-3
10^-4
10^-5
-10
0
10
20
30
40
50


Effect of packet aggeregation on MC-MIMO system performance
SNR vs BER
PktS Var Lat0
PktS Var Lat1
PktS Var Lat2
PktS 8448
BER
SNR (dB)
10^0
10^-1
10^-2
10^-3
10^-4
10^-5
10^-6
-10
0
10
20
30
40
50

Contributions

- Characterised the 5G LDPC codes.
- Explained the difference between Realtime and Non-Realtime systems.
- Investigated the performance of video transmission using LDPC.
- Shown the degradation of performance of Realtime systems.
- Proposed a novel aggregation policy to combat the performance degradation.
- Thoroughly characterised the underlying Latency-Reliability trade off and its dependence on the system SNR.
- Explicated the effect of the aggregation policy on MC-MIMO system.
- Demonstrated the transmission of H.264 encoded video over the MC-MIMO system.

Future Scope

- Performance investigation of LDPC codes close to their Error floor (i.e. BER < 10^{-10}).

References

- [1] Jagyasi, A. Daher, and M. Coupechoux, “Multi-cell mimo transceiver design for mission-critical communication,” 2019.
- [2] N. A. Mohammed, A. M. Mansoor, and R. B. Ahmad, “Mission-critical machine-type communications: An overview and perspectives towards 5g,” IEEE Access, 2019.
- [3 Tom Richardson and Ruediger Urbanke. Modern coding theory. Cambridge university press, 2008.
- [4] Stefan Parkvall, Erik Dahlman, Anders Furuskar, and Mattias Frenne. Nr: The new 5g radio access technology. IEEE Communications Standards Magazine, 1(4):24–30, 2017.
- [5] Khaled Salah, Fahd Al-Haidari, MH Omar, and A Chaudhry. Statistical analysis of h. 264 video frame size distribution. IET communications, 5(14):1978–1986, 2011.